

Mawlana Bhashani Science and Technology University
Department of Information and communication Technology (ICT)
3rd Year 2nd Semester B.Sc. (Engg.) Final Examination-2024

Course Title: Computer Networks

Course Code: ICT-3201

Time: 3 Hours

Marks: 70

N.B.:

i) Answer any five of the following questions

ii) The sequence should be maintained in answering each of the questions

1. a) Explain why computer networks is necessary for an undergraduate student of ICT department. 3
- b) Compare and contrast between the OSI and TCP/IP reference model. 4
- c) Apply OSI data transmission terminologies to explain how an email message is transmitted over a network. 4
- d) Explain the principle of protocol layering for network model. 3
2. a) Explain how the end-to-end principle and the "IP over everything" principle together form the foundation of the Internet's design. Give a real-world example where these principles allow for easy adoption of new network technologies. 4
- b) Describe how protocol encapsulation and multiplexing/demultiplexing work together in the TCP/IP model. Use the example of an HTTP request being sent over Ethernet to illustrate the process. 5
- c) The TCP/IP model is often compared to the OSI model. In a single, clearly labeled diagram, map the four layers of the TCP/IP model to the seven layers of the OSI model, showing their correspondence. 5
3. a) Subnet the network 172.16.0.0/16 to support at least 500 hosts per subnet. Determine the number of subnets and subnet mask. 3
- b) Compare and analysis distance vector and link state routing algorithms 3
- c) Calculate the routing table for any network with 6 nodes and 10 edges using the distance vector routing algorithm. 4
- d) A network has the IP address 192.168.10.0/26. Now find 4
 - i. Subnet Mask
 - ii. Total number of sub-networks
 - iii. Total number of IP address on each sub-network
 - iv. Total number of host on each sub-network
 - v. Ranging of host address on each sub-network
4. a) Analyze the function of ICMPv4 in the IP network layer. Explain with examples how ICMP error-reporting messages (such as Destination Unreachable and Time Exceeded) assist in network diagnostics and troubleshooting. 5
- b) Describe how FTP uses out-of-band control and data connections (ports 20 and 21). Discuss the advantages and disadvantages of this approach compared to using a single connection for both control and data, as in HTTP. 5
- c) Explain the process of Address Resolution Protocol (ARP) and its role in enabling communication between devices on the same broadcast domain. Discuss why ARP caching is important for network efficiency. 4

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5. a) Compare the Stop-and-Wait and Sliding Window flow control mechanisms in terms of efficiency, resource utilization, and suitability for different network conditions. Provide a scenario where each method would be preferred. 5
 - b) Explain how Automatic Repeat Request (ARQ) mechanisms handle error control in the data-link layer. Differentiate between Stop-and-Wait ARQ, Go Back-N ARQ, and Selective Repeat ARQ in terms of retransmission strategy and buffer requirements. 5
 - c) Illustrate with a diagram or example how the Sliding Window protocol can be combined with ARQ to achieve both flow control and error control simultaneously. Explain how window size affects performance. 4
 6. a) Compare and contrast Single-bit, Multiple-bit, and Burst errors in terms of their causes, detection difficulty, and impact on data integrity. Provide an example scenario where each type might occur in a communication system. 5
 - b) Describe the Cyclic Redundancy Check (CRC) method for error detection. Explain the role of the polynomial divisor, codeword formation, and the division process at both sender and receiver. Why is CRC more robust than parity check? 5
 - c) A company is transmitting the dataword 101110101011 over a communication channel using the CRC error-detection method. The chosen divisor for CRC is 110101. The transmission system appends the appropriate number of zeros to the dataword, calculates the CRC remainder, and sends the complete frame over the channel. During transmission, a single-bit error occurs in the fifth bit of the transmitted frame. Determine the CRC code, construct the final transmitted frame and explain whether the CRC method can detect the single-bit error in this situation. 4
 7. a) Explain the role of DNS in network addressing. Describe the process through which a host obtains the IP address of a remote host using DNS and how this address is used in subsequent packet forwarding. 5
 - b) Compare and contrast DNS and HTTP in terms of their transport layer protocols (UDP vs. TCP), communication models, and roles in the Internet infrastructure. Explain why DNS typically uses UDP despite being a critical service. 5
 - c) Explain the process of email transmission using SMTP, POP3, and IMAP. Discuss the roles of User Agents, Message Transfer Agents, and mail servers, and analyze the differences between POP3's delete and keep modes. 4
 8. a) Explain the role of TCP flags (SYN, ACK, FIN, RST, PSH, URG) in connection management, data transfer, and error handling. Illustrate how these flags are used in a typical TCP three-way handshake and connection termination. 5
 - b) Analyze how TCP ensures reliable data delivery through sequence numbers, acknowledgements, and retransmission timers. Discuss what happens when segments arrive out of order or are lost, and how duplicate segments are handled. 4
 - c) Compare the functions of the four TCP timers: Keep-alive, Retransmission, Persist, and Timed-Wait. Explain how each timer contributes to connection stability, flow control, and crash recovery. 5